

Bugs found: 4

1. Static collection 'leak' grows indefinitely, causing a potential memory leak.
2. Attempting to add an element to a fixed-size list returned by `Arrays.asList()` results in `UnsupportedOperationException`.
3. Modifying a collection while iterating over it via an enhanced for-loop results in `ConcurrentModificationException`.
4. Explicitly casting an `Integer` object to a `String` results in a `ClassCastException`.